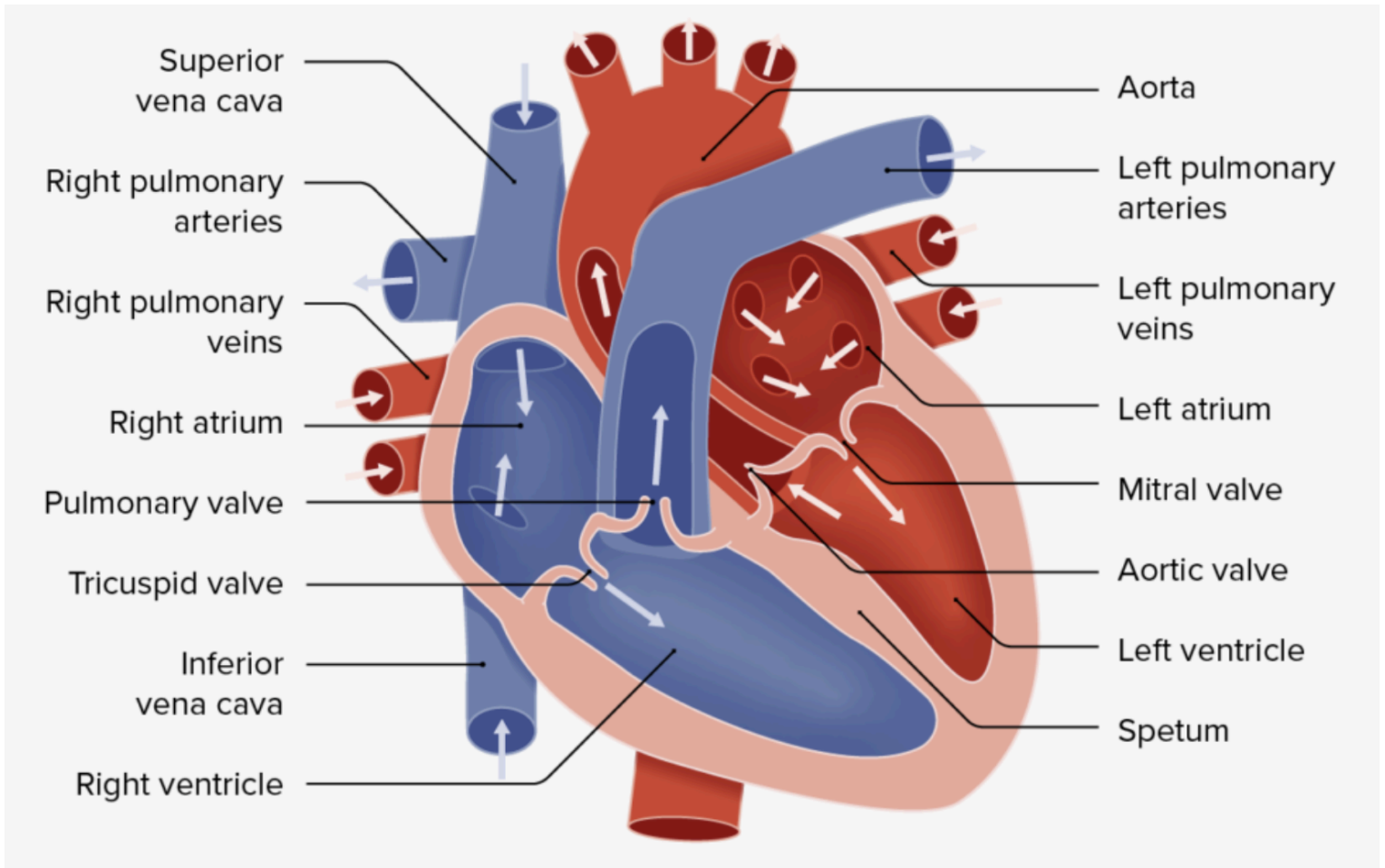




Ischemic Heart Disease (IHD)

- Leading cause of death worldwide
- Over 30% of Americans over 35 die from IHD
- Caused by reduced blood flow to the heart muscle (myocardium)

Ischemia

- Not just a drop in oxygen supply (hypoxia), but also a decrease in nutrient delivery and waste removal due to reduced blood flow

Coronary Arteries

- **Left Coronary Artery (LCA):** Branches into the left circumflex artery (LCX) and the left anterior descending artery (LAD)
- **Right Coronary Artery (RCA):** Includes branches such as the posterior descending artery (PDA)

Blood Supply Areas

- **LAD:** Supplies 2/3 of the interventricular septum and the anterior wall of the left ventricle.
- **LCX:** Supplies the lateral wall of the left ventricle.
- **RCA and PDA:** Supply the right ventricular wall and part of the posterior wall of the septum, as well as the posterior inferior wall of the left ventricle

Causes of Ischemic Heart Disease

- **Atherosclerosis:** The primary cause, responsible for over 90% of cases. Involves the formation of lipid-based plaques under the endothelium, creating blockages.
 - **Stable Plaques:** Have a cap and cause stable angina, a type of chest pain that resolves with rest or vasodilators.
 - **Unstable Plaques:** Can rupture, leading to platelet accumulation and thrombus formation, causing acute coronary syndromes like unstable angina and MI.

Types of Angina

- **Stable Angina:** Caused by stable plaques. Chest pain resolves with rest or vasodilators (Nitroglycerine)
- **Unstable Angina:** Caused by unstable plaques. Chest pain may not resolve with rest or vasodilators and can progress to MI.
- **Prinzmetal or Variant Angina:** Caused by transient vasospasm. Chest pain resolves with vasodilators.

Acute Coronary Syndrome (ACS)

- Includes unstable angina and myocardial infarction (MI).
- **NSTEMI (Non-ST Elevated Myocardial Infarction):** Involves subendocardial infarction.
- **STEMI (ST Elevated Myocardial Infarction):** Involves transmural infarction.

Diagnosing MI with ECG

- **ECG Leads and Infarction Location:**
 - **V1 to V4:** Anterior wall infarction (LAD occlusion).
 - **V5, V6, AVL, and lead I:** Lateral wall infarction (left circumflex artery).
 - **Leads III, AVF, and II:** Inferior wall infarction (RCA or PDA).

ECG Changes in MI

- **NSTEMI:** ST depression. non-ST-elevation myocardial infarction
- **STEMI:** ST elevation. ST-elevation myocardial infarction

Troponin

- Elevated for more than **0.04 troponin** levels indicate **myocardial cell death**, as troponin is released when the myocardium starts to die

Troponin I - Normal Range: Less than 0.04 ng/mL (nanograms per milliliter)

Troponin T - Normal Range: Less than 0.01 ng/mL

Other Causes of Ischemic Heart Disease

- **Vasospasm:** Can cause transient or severe, long-term vasospasm leading to MI.
- **Microvascular Dysfunction:** Affects smaller blood vessels, potentially leading to conditions like Takotsubo myopathy (broken heart syndrome).

Myocardial Infarction

Causes

- **Hyperlipidemia:** Leads to **atherosclerosis**, with fats depositing on the walls of arteries.
- **Deep Venous Thrombosis (Thrombophlebitis)**
- **Diabetes Mellitus**
- **Metabolic Syndrome:** Characterized by a high BMI.
- **Obesity**
- **Hypertension**

Symptoms of MI

- Chest pain (squeezing, choking chest pain)
- Shortness of breath (SOB)
- Dyspnea
- Hypoxia
- Nausea
- Vomiting
- Epigastric pain

Nitroglycerine Precautions

- Always keep nitroglycerine in a black bottle in a dark place; do not store it in the car.
- Common side effect: headache (an expected effect).
- As a vasodilator, nitroglycerine can cause hypotension, leading to orthostatic hypotension.
- It is recommended to lie down when taking nitroglycerine sublingually.
- Forms: sublingual tablets, patch (apply with gloves), cream (apply on the arm), and IV.
- Always monitor blood pressure when taking nitroglycerine.

Myocardial Infarction (MI)

- MI is an acute condition resulting in irreversible necrosis of the heart muscle.
- After an MI, it is crucial to monitor the patient to prevent another MI.

Stable Angina

- Stable angina can be a chronic condition.
- Recommendations: stop smoking, lose weight, avoid fatty foods and alcohol, and use nitroglycerine to control chest pain.
- Triggers: exercise, physical work, unhealthy food, alcohol, and immobilization.
- Chest pain protocol:
 - Take the first dose of nitroglycerine while lying down. (If the pain does not subside after the first dose, call 911. The second and third doses provide an additional 10 minutes for emergency services to arrive)
 - If pain persists after 5 minutes, take a second dose.
 - If pain still persists, take a third dose (maximum of 3 doses, 5 minutes apart).
- Signs of **stable angina** turning **unstable**: nausea, vomiting, epigastric pain, and dyspepsia (early symptoms of MI, not to be confused with gastrointestinal issues).

Hospital Protocol for Suspected MI

1. Initial Actions:

- Connect the patient to a peripheral device and EKG (to check for STEMI or NSTEMI).
- Draw blood for lab tests: check D-Dimer, aPTT (coagulation panel), and **troponin** levels (troponin levels above **0.04 indicate MI**. **Troponin I - Normal Range:** Less than 0.04 ng/mL **Troponin T - Normal Range:** Less than 0.01 ng/mL)
 - a) **Coagulation panel**
 - i) **Prothrombin Time (PT):** Measures the time it takes for blood to clot. It evaluates the extrinsic pathway and common pathway of coagulation.
 - ii) **International Normalized Ratio (INR):** Standardizes the PT results regardless of testing methods, making it easier to compare results across different labs.
 - iii) **Activated Partial Thromboplastin Time (aPTT):** Measures the time it takes for blood to clot, focusing on the intrinsic pathway and common pathway of coagulation.
 - iv) **Thrombin Time (TT):** Measures the time it takes for a clot to form in the presence of thrombin, evaluating the final step of the coagulation process.
 - v) **Fibrinogen Level:** Assesses the amount of fibrinogen, a protein essential for blood clot formation.
 - vi) **D-Dimer:** A test that helps diagnose clotting disorders, such as deep vein thrombosis (DVT) and pulmonary embolism, by measuring the presence of fibrin degradation products.
- Look for ST segment elevation

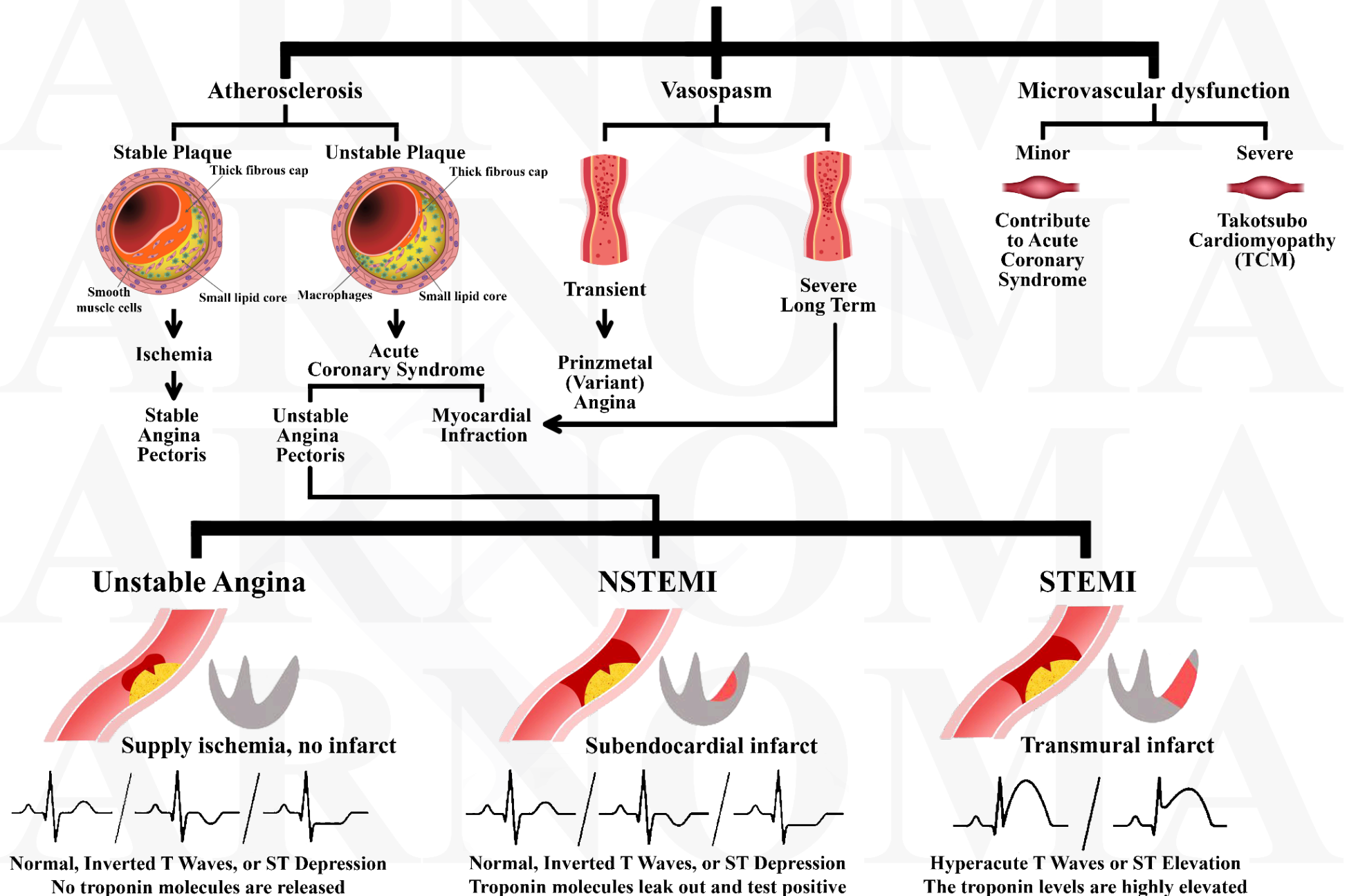
2. First Treatment:

- Administer Aspirin (**suspected** MI patients should take an emergency chewable Aspirin 160 mg - 2 baby doses)

MONA Protocol for MI

- **Morphine:** For severe chest pain radiating to the jaw, left shoulder, back, and epigastric area. Assess the severity of the pain (MI pain scale is 8-10; pain less than 5 is unlikely to be MI).
- **Oxygen:** For shortness of breath (SOB) and hypoxia.
- **Nitroglycerine:** To dilate blood vessels.
- **Aspirin:** To reduce the risk of thrombus formation.
- **Additional medications:** Heparin (anticoagulant) or Enoxaparin (to prevent further clot damage).

Ischemic Heart Disease



Stable Angina

Condition/Cause	Treatment	Monitor
Coronary arteries are narrowed by plaque buildup with a cap. During physical activity or stress, the heart needs more oxygen-rich blood than the narrowed arteries can supply, causing chest pain. Not just a drop in oxygen supply but also a decrease in nutrient delivery and waste removal due to reduced blood flow	Chest pain protocol: 1. Take the first dose of nitroglycerine while lying down. (If the pain does not subside after the first dose, call 911. The second and third doses provide an additional 10 minutes for emergency services to arrive) 2. If pain persists after 5 minutes, take a second dose. 3. If pain still persists, take a third dose (maximum of 3 doses, 5 minutes apart). Signs of stable angina turning unstable: nausea, vomiting, epigastric pain, and dyspepsia (indigestion) (early symptoms of MI, not to be confused with gastrointestinal issues).	Complications Unstable Angina and MI Differentiate • Predictable, triggered by exertion or stress. • Relieved by rest or nitroglycerin. • Caused by fixed atherosclerotic plaque. • Pain usually lasts for a few minutes (typically less than 10 minutes)
Symptoms Symptoms of stable angina are predictable and occur in a pattern, often triggered by similar activities and relieved by rest or medication. Chest Pain or Discomfort: • Typically feels like pressure, squeezing, or fullness in the chest. • Often triggered by physical exertion, stress, or cold weather. • Usually lasts a few minutes and goes away with rest or medication (such as nitroglycerin). Pain Radiating to Other Areas: Pain can spread to the arms, neck, jaw, shoulder, or back. Pain usually lasts for a few minutes (typically less than 10 minutes). Shortness of Breath: Difficulty breathing during physical activity or stress. Fatigue: Feeling unusually tired during or after physical activity. Nausea: Sometimes accompanied by feelings of nausea or lightheadedness. Sweating: Cold, clammy skin.	Note	

Unstable Angina NSTEMI & STEMI

Condition/Cause	Treatment	Monitor
A plaque in a coronary artery ruptures or forms a blood clot, partially blocking the artery. This significantly reduces blood flow to the heart. See photo on page 74	Initial Actions: - Connect the patient to a peripheral device and EKG (to check for STEMI or NSTEMI). - Draw blood for lab tests: check D-Dimer, aPTT (coagulation panel), and troponin levels (troponin levels above 0.04 indicate MI. Troponin I - Normal Range: Less than 0.04 ng/mL Troponin T - Normal Range: Less than 0.01 ng/mL) First Treatment: - Administer Aspirin (suspected MI patients should take an emergency chewable Aspirin 160 mg - 2 baby doses) MONA Protocol for MI M - Morphine O - Oxygen N - Nitroglycerine A - Aspirin	- EKG - HR - BP - Troponin
Symptoms Unstable angina is a medical emergency and indicates that MI could occur soon. Chest Pain or Discomfort: - Sudden onset and may occur at rest or with minimal exertion. - Pain in the chest: Feels like pressure, squeezing, or severe. - May be more intense and longer-lasting than stable angina. - Pain can spread to the arms, neck, jaw, shoulder, or back, similar to stable angina but often more severe. - Difficulty breathing even without physical activity. - Nausea, Feeling sick to the stomach, which can be accompanied by vomiting. - Profuse sweating, often cold and clammy. - Unusual tiredness without a clear cause. - Feeling faint or unsteady. - Symptoms may not fully resolve with rest or the use of nitroglycerin.		Complications Myocardial Infarction Differentiate - Sudden onset can occur at rest. - More severe and longer-lasting - May not be relieved by rest or nitroglycerin. - Caused by rupture of an atherosclerotic plaque. - Pain can last longer (up to 20 minutes or more).

Variant Angina / Prinzmetal's angina

Condition/Cause	Treatment	Monitor
Coronary arteries experience temporary spasms, narrowing the arteries. These spasms restrict blood flow to the heart, causing chest pain, typically at rest and often during early morning.	Note	Complications
Symptoms Chest Pain: Occurs at rest, often during the night or early morning. Can be severe and feels like pressure, squeezing, or tightness in the chest. Pain Radiating to Other Areas: Pain can spread to the arms, neck, jaw, shoulder, or back. Pain typically lasts for a few minutes. Shortness of Breath: Difficulty breathing during an angina attack. Nausea: Feeling sick to the stomach, which may be accompanied by vomiting. Sweating: Cold, clammy skin. Palpitations: Feeling of a rapid or irregular heartbeat.		Differentiate • Occurs at rest, often during the night or early morning. • Caused by coronary artery spasm. • Not typically triggered by exertion or stress. • Responds to calcium channel blockers and nitrates. • Pain typically lasts for a few minutes.

